

The Workload Class Problem: Coherence-SLA Classification as a First-Class Scheduling Primitive at the Tactical Edge

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Abstract

The preceding four Wave 1 papers have specified the tactical substrate's mechanical components: DTN custody transfer (P3), WAL + CRDT state coherence (P4), the AI Supervisor out-of-path control plane (P5), and governed dynamic protocol selection (P6). Each of these components requires a key input that none of them generates: the workload class of the data being processed. P3's custody scheduler prioritizes bundles by workload class. P4's CRDT merge engine escalates or resolves conflicts based on workload class. P5's Supervisor applies autonomy envelopes that are workload-class-dependent. P6's protocol selection decision matrix uses workload class requirements as its primary constraint input. The workload class is the common scheduling primitive that makes the four substrate components governable rather than technically capable without direction.

This paper argues that the workload class problem — how to define, assign, enforce, and evolve a taxonomy of operational data types that the substrate can use as a scheduling primitive — is not a data modeling problem but a governance architecture problem. The taxonomy is not derivable from the technical properties of the data; it is an encoding of operational doctrine and coherence requirements that only the H half can supply. A workload class taxonomy authored by an engineering team reflects engineering priorities; a workload class taxonomy authored by the operational community reflects operational doctrine. The substrate enforces whichever taxonomy it receives; its enforcement does not validate the taxonomy's operational correctness — only governance review can do that.

The paper develops a four-dimension workload class framework: coherence class (what consistency guarantee does this data type require across disconnected nodes?), SLA class (what latency bound and delivery confirmation does this data type require?), security class (what authentication and encryption profile does this data type require?), and authority class (what level of governance oversight does the processing of this data type require?). Each dimension is independently assigned per data type; the combination of the four dimensions produces the complete workload class specification that the substrate uses as its scheduling primitive. The §6 Governor Application specifies how the HGC³AE² framework governs the taxonomy's definition, maintenance, and enforcement.

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ABSTRACT

Workloads on the tactical substrate (EDO-P1) do not share a uniform service-level objective. Some require strict coherence with the authoritative tier (mission-critical governance state); some tolerate eventual consistency (telemetry aggregation); some operate on episodically-consistent state (work-in-flight). This paper specifies a **Coherence-SLA classification** that names the workload's coherence requirement explicitly and routes the workload to the substrate mode that satisfies it.¹² The classification is the operational form of HGC³AE²'s **C³** (Coherence) under DDIL conditions: coherence is a workload property, not a substrate property, and the architecture must route by class.

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1. INTRODUCTION

Cloud-native workload classification typically treats all workloads under a single consistency model (strong consistency in databases; eventual consistency in messaging; cache semantics in CDNs).³ The tactical substrate (EDO-P1) cannot satisfy a single consistency model across all its workloads simultaneously; partition forces a choice (CAP theorem),⁴ and the choice cannot be uniform because workloads have heterogeneous coherence requirements.

The architectural commitment of this paper is to classify workloads explicitly by coherence-SLA: each workload declares its required coherence class, the substrate routes accordingly, and the federation envelope enforces the routing. Coherence becomes a property the agentic operating model can reason about per-workload, not a property the substrate provides uniformly.

§2 names five failure modes of unclassified workload deployment. §3 specifies the Coherence-SLA classification. §4 treats classification as continuous. §5 binds to Skipjack. §6 returns to HGC³AE².

2. FAILURE MODES OF UNCLASSIFIED WORKLOAD DEPLOYMENT

Over-engineered consistency. Workloads that tolerate eventual consistency are routed to strict-consistency substrate, paying latency and availability cost without coherence benefit.

Under-engineered consistency. Workloads requiring strict coherence are routed to eventual-consistency substrate, accumulating drift that the workload cannot tolerate.

1 Justin H. Kuiper, CISSP, *HGC³AE² Framework* (v1.0-preprint, 2026), zenodo.org/records/19869285.

2 Betsy Beyer et al., *Site Reliability Engineering: How Google Runs Production Systems* (Sebastopol, CA: O'Reilly, 2016).

3 Martin Kleppmann, *Designing Data-Intensive Applications* (Sebastopol, CA: O'Reilly, 2017).

4 Eric A. Brewer, "CAP Twelve Years Later: How the 'Rules' Have Changed," *IEEE Computer* 45, no. 2 (2012): 23–29, <https://doi.org/10.1109/MC.2012.37>.



SLA-substrate mismatch surprise. Workload class is discovered at failure time rather than admission time; substrate-level recovery cannot compensate for design-level mismatch.

Partition-policy ambiguity. When partition forces a CAP choice, single-policy substrates produce the wrong answer for half the workload mix.

Classification erosion. Workloads classified once but never re-evaluated drift away from their classification as the workload itself evolves.

3. THE COHERENCE-SLA CLASSIFICATION

Five workload classes are load-bearing for agentic operation on the tactical substrate.

Strict-coherence governance. Envelope updates, policy authority changes, governance event log writes. Routes only to authoritative substrate; partitions halt the workload until reachable.

Eventual-coherence telemetry. Observability metrics, audit trails, non-decision-critical state. Routes to any substrate; merge resolves divergence on reconciliation.

Episodic-coherence work-in-flight. Agentic work that runs between reconciliation checkpoints. Routes to any substrate with custody-transfer discipline (EDO-P3); state coherence resolves at next reconciliation.⁵

Local-only operational state. State that is purely local to a substrate (e.g., transient routing decisions). No federation-level coherence requirement; routes locally and is not merged.

Strict-causality persistence. State whose causal ordering matters (e.g., financial transactions, governance authorizations). Requires WAL-discipline persistence (EDO-P4) and explicit causal-ordering enforcement.⁶⁷

The classification is governance-defined at workload-admission time; the multi-policy router (MAO-P2) reads the classification and routes accordingly.⁸

4. CLASSIFICATION AS CONTINUOUS PROCESS

Three phases. **Admission classification** at workload entry: explicit Coherence-SLA assignment. **Runtime adherence** at every state-modifying event: the substrate enforces the classification. **Re-classification review** at sprint cadence: workloads whose nature has evolved are re-classified at governance ceremony.

The continuous-process framing prevents classification erosion: workloads that change over operating time are re-classified rather than left at stale classifications.

5. SKIPJACK COUPLING

Skipjack mechanisms operate over each Coherence-SLA class with class-appropriate enforcement.⁹ **Context integrity** is enforced per-class (strict for governance, eventual for telemetry). **Structured routing** uses classification as a routing-decision input. **HITL control**

5 EDO-P3 promotion (in concurrent drafting; yks-build#95).

6 EDO-P4 promotion (in concurrent drafting; yks-build#96).

7 Colin J. Fidge, "Timestamps in Message-Passing Systems That Preserve the Partial Ordering," *11th Australian Computer Science Conference* 10 (1988): 56–66.

8 MAO-P2 *Heterogeneous Compute Routing* (in concurrent drafting; yks-build#93).



points fire on classification anomalies (a workload exhibiting behavior inconsistent with its class). **Iterative feedback** closes via re-classification at sprint review.

6. IMPLICATIONS AND CONCLUSION

The implications extend to architecture (classification is a first-class admission decision), tooling (the multi-policy router must consume classification), operational discipline (classification reviews are scheduled, not ad-hoc).

The core claim: **coherence is a workload property requiring explicit classification; uniform substrate-level consistency cannot satisfy heterogeneous coherence requirements on the tactical substrate.**

Returned to HGC³AE² letter by letter:

H — Human-governed. Classification policy is governance-authored.

G — Curated Context. The classification registry is curated context.

C¹ — Cybersecurity. Classification is itself a security-significant property (mis-classification can produce security failures).

C² — Continuity. Continuity per-class — strict for governance, episodic for work-in-flight.

C³ — Coherence. This paper's load-bearing commitment: coherence as workload property.

A — Agentically Engineered. Classification consumed by routing; routing is agentic.

E — Executed. Continuous classification enforcement.

The exponent — the closure — is the classify-route-review loop.

The remaining EDO papers (classification, accreditation, interoperability, deployment) operate on the workload-class substrate this paper specifies.¹⁰¹¹

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9 Justin H. Kuiper, CISSP, *Skipjack Protocol* (v1.0-preprint, 2026), zenodo.org/records/19869293.

10 EDO-P8 *Classification Problem* promotion (in concurrent drafting; yks-build#107).

11 EDO-P11 CAPSTONE promotion (in concurrent drafting; yks-build#110).